

Literature Status Note: Uniform Algebras and the Complex Mazur–Ulam Property

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Status. `literature_implied_answer` (**partial subcase**). The complex-target reading of the uniform-algebra Mazur–Ulam conjecture in arXiv:2101.01994 is answered positively by the later paper arXiv:2107.01515. The full real-target Mazur–Ulam property remains open in the supporting paper.

Original source. O. Hatori, S. Oi, and R. Shindo Togashi, *Tingley’s problems on uniform algebras*, arXiv:2101.01994. The source proves that a surjective isometry between the unit spheres of two uniform algebras extends to a surjective real-linear isometry. In Section 8 (“Remarks”), the authors then conjecture that a uniform algebra satisfies the Mazur–Ulam property, meaning that the target should be allowed to be an arbitrary Banach space rather than another uniform algebra.

Supporting source. O. Hatori, *The Mazur–Ulam property for uniform algebras*, arXiv:2107.01515. The introduction explicitly recalls the arXiv:2101.01994 theorem and states that the new paper proves the complex Mazur–Ulam property for uniform algebras. The decisive statement is Theorem 4.2: every uniform algebra satisfies the complex Mazur–Ulam property. Equivalently, if A is a uniform algebra and $T : S(A) \rightarrow S(E)$ is a surjective isometry onto the unit sphere of an arbitrary complex Banach space E , then the homogeneous extension of T is a surjective real-linear isometry $A \rightarrow E$.

Identification. This is an implication rather than a same-paper answer. The original paper asked for the Mazur–Ulam property after proving the uniform-algebra-to-uniform-algebra Tingley theorem. The later paper proves exactly the arbitrary-complex-target version. Thus the natural complex Banach-space form of the conjecture is settled positively in the literature.

Remaining scope. The later paper does not claim the full real-target Mazur–Ulam property. Indeed its final remarks say that it merely proves the complex Mazur–Ulam property and still conjectures the Mazur–Ulam property for uniform algebras, more generally for closed subalgebras of $C_0(Y, \mathbb{C})$.

There is also a visible reason the same proof does not directly cover the real form of a complex uniform algebra. In Hatori’s real-space sufficient condition one needs

$$M_{p,\alpha} = \{a \in S(B) : p(a) = \alpha\}$$

for real face representatives p and all $-1 \leq \alpha \leq 1$. For the real Banach space underlying a complex uniform algebra, the natural real face functionals are $p = \operatorname{Re}(\gamma\delta_x)$. Taking $p = \operatorname{Re}\delta_x$ and a sphere

element f with $f(x) = i$, one has $p(f) = 0$, but

$$\text{dist}(f, \{g \in S(A) : g(x) = 1\}) \geq |i - 1| = \sqrt{2} > 1.$$

Hence $f \notin M_{p,0}$. The real criterion in arXiv:2107.01515 therefore does not automatically settle the remaining real-target conjecture.

Evidence checked. Local arXiv sources and PDFs were checked for arXiv:2101.01994 and arXiv:2107.01515. Cheap run indexes showed no prior exact packet for arXiv:2101.01994. Web search for the source title together with “Mazur–Ulam property” led to arXiv:2107.01515, whose title and theorem statements match the complex-target subcase.